SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS: **Total Marks: 20**

1: HTTP Response Analysis

A web server returns the following HTTP response header:

HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html Content-
Length: -150 Connection: keep-
alive

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with Content-Lengthand explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: **DOM Structure Analysis**

Consider the following DOM tree representation:

```
<html>
  <head>
  <title>TestPage</title>
</head>
  <body>
  <div>
  <p>HelloWorld
  </div>
 </body>
 </html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the <img>tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

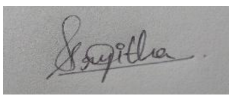
1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

Code of Conduct:

I s.srijitha(192471024(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Question 1: HTTP Response Analysis
1. Identify any incorrect or invalid fields in the header.
- **Content-Length: -150** is invalid because the value cannot be negative.
- The remaining fields (HTTP/1.1, Date, Content-Type, and Connection) are valid.

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2. Analyze the issue with Content-Length and explain its impact.
- **Content-Length** specifies the size of the response body in bytes.
- A negative value is invalid.
- The browser cannot determine the correct amount of data to receive.
- This may cause page loading errors or incomplete content.

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3. Determine whether the status code matches the response correctness.
- **200 OK** means the request was successful.
- However, since the **Content-Length** is invalid, the response is not completely correct.
- The status code should only be **200 OK** if the response header is valid.

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4. Suggest corrections for the identified errors.
Correct Header:
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: 150
Connection: keep-alive
- Replace the negative Content-Length with the correct positive value.

Question 2: DOM Structure Analysis
Given Code
```
<html>
<head>
<title>TestPage</title>
</head>
<body>
<div>
<p>HelloWorld
</div>
</body>
</html>
```
1. Identify missing or improperly closed elements.
- Missing </p> closing tag.
- Missing </div> before closing the paragraph properly.
- Missing <!DOCTYPE html>.
- Missing lang="en" attribute in the <html> tag.

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2. Analyze how the browser will interpret this structure.
- Browsers automatically close the <p> tag when necessary.
- The page may still display correctly.
- Different browsers may interpret the HTML differently.

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3. Determine the impact on rendering and layout.
- Most browsers will display the text normally.
- Invalid HTML may lead to inconsistent layouts.
- Accessibility and HTML validation are affected.

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4. Suggest corrections to make the DOM valid.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>TestPage</title>
</head>

<body>

<div>
<p>Hello World</p>
</div>

</body>
</html>
```

Question 3: GET vs POST in Login System

Method	URL Example	Data Visibility	Security
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in Request Body	Moderate

---

1. Interpret the differences in data transmission between GET and POST.

GET

- Sends data through the URL.
- Data is visible in the browser.
- Used for retrieving information.

POST

- Sends data in the request body.
- Data is not shown in the URL.
- Used for submitting sensitive information.

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2. Analyze which method is more suitable for login and why.

- **POST** is more suitable.
- Login credentials are sent in the request body.
- Passwords are not exposed in the URL.
- It provides better security than GET.

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3. Identify security risks in using GET for authentication.

- Username and password appear in the URL.
- URLs may be stored in browser history.
- Sensitive information can be logged by servers.
- Passwords may be exposed if the URL is shared.

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4. Suggest best practices for secure login implementation.

- Use **POST** instead of GET.
- Use **HTTPS** for encrypted communication.
- Validate input on both client and server.
- Store passwords using secure hashing.
- Use secure session management.

Question 4: HTML Code Inspection
Given Code

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
</html>
```

1. Identify missing closing tags.

Missing closing tags:

- </h1>
- </p>

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2. Analyze issues with the <img> tag.

- The <img> tag does not have an **alt** attribute.
- The **alt** attribute is important for accessibility.
- It helps screen readers and displays text if the image cannot load.

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3. Determine how browsers will handle these errors.

- Browsers automatically close missing tags.
- The page may still display correctly.
- However, the HTML is invalid and may behave differently across browsers.

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4. Suggest a corrected version of the code.

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<title>Sample</title>
</head>

<body>

<h1>Welcome</h1>

<p>This is a paragraph.</p>

<img src="image.jpg" alt="Sample Image">

</body>
</html>
```

Question 5: Browser-Server Communication Flow
Given Flow

1. User enters a URL.
2. DNS lookup occurs.
3. HTTP request is sent.
4. Server processes the request.
5. Server returns the response.
6. Browser renders the page.

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1. Interpret each step in the communication flow.
Step 1: User enters a URL in the browser.
The browser receives the website address.
Step 2: DNS lookup occurs.
The domain name is converted into an IP address.
Step 3: HTTP request is sent.
The browser sends an HTTP request to the web server.
Step 4: Server processes the request.
The server processes the request and prepares the response.
Step 5: Response is returned.
The server sends HTML, CSS, JavaScript, and other resources back to the browser.
Step 6: Browser renders the page.
The browser displays the webpage to the user.

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2. Identify where delays can occur.

- Slow DNS lookup.
- Slow internet connection.
- High server processing time.
- Large webpage files.
- Slow browser rendering.

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3. Analyze the role of DNS in the process.

- DNS converts the domain name into an IP address.
- Without DNS, the browser cannot locate the web server.
- It acts like the internet's address book.

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4. Suggest optimizations to improve performance.

- Use DNS caching.
- Enable browser caching.
- Compress files using Gzip or Brotli.
- Minify HTML, CSS, and JavaScript.
- Use a Content Delivery Network (CDN).
- Optimize images and reduce file sizes.
